

Money and Inflation

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Overview

1. The definition of money.
2. How the money supply is created.
3. The relationship between money growth and inflation in the long run.
 - Quantity theory of money.
4. The relationship between inflation and interest rates.
 - Fisher equation and Fisher effect.
5. Money demand
6. The money market in the long run
7. How does monetary policy impact expected inflation?
8. Why is inflation costly?

What is money?

- **Money**: An asset that can be used without cost to perform exchanges for goods and services.
- **Money supply**: The quantity of money in circulation in the economy as a given point in time $\implies$ stock variable.
- **Monetary policy**: Performed by the Central bank to change the money supply and the interest rate to attain its macroeconomic objectives
 - Price stability
 - Macroeconomic stabilization

What is money?

1. Medium of exchange.

- We can use it to buy goods and services.
- The ease with which an asset can be exchanged for goods and services is called the **liquidity** of an asset.
- Money is the most liquid asset.

2. Store of value.

- If we earn \$5 today, we will still have \$5 tomorrow.

3. Unit of account.

- Prices are denominated in monetary values.

Types of money

- **Commodity money**
 - Objects with intrinsic value.
 - Silver or gold coins, etc.
- **Commodity-backed money**
 - Paper money that can be exchanged for a specific commodity.
- **Fiat money**
 - Has no intrinsic value.
 - Paper money (legal tender currency).

Monetary aggregates

- Money can be defined more or less broadly
- Money supply includes:
 - M_0 = Cash and coins (currency, C).
 - $M_1 = M_0 + \text{demand deposits.}$
 - $M_2 = M_1 + \text{time deposits.}$
- There are further measures that increase in size including other liquid financial assets (bonds). $M_3, M_4, \dots$
- There is no consensus on which measurement is best to use but typically you will see M_1 or M_2 .

Creation of money: Definitions

- **Reserves (R)**: Currency banks hold

$$R = \text{Required reserves (RR)} + \text{Excess reserves (ER)}$$

- **Deposits (D)**: Money people store in banks (current accounts, savings etc).
- **Monetary Base (B)** = Currency + Reserves

What the Central bank controls.

- **Monetary Supply (M)** = Currency + Deposits

Influenced by the Central bank but also depends on banks, and household savings

Creation of money: Monetary base

The central bank controls the monetary base in the economy through:

1. Open-market operations - the purchase and sale of government bonds.
 - Buy government bonds $\implies$ Money supply $\uparrow$
 - Sell government bonds $\implies$ Money supply $\downarrow$
2. Refinancing operations with banks (reverses)
3. Buying and selling foreign and domestic currency on international markets

In modern economies 2 is the most important.

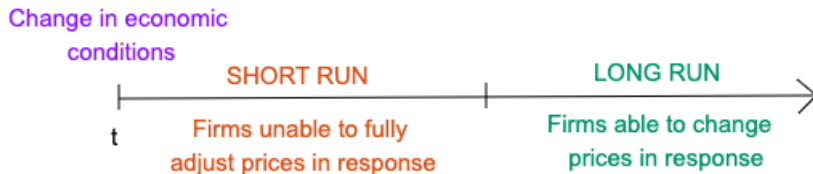
Creation of money: Monetary supply

- Notice the most commonly used measures of money go beyond currency and reserves.
- Q: How is money created?
 - ⇒ The Central bank operations on the monetary base.
 - ⇒ Through commercial banks.
 - ⇒ Through individual saving decisions.

Money and inflation in the long run: Quantity Theory of Money

- Mainly due to **David Hume** (Scottish Enlightenment philosopher, 1711-1776).
 - Introduced a theory to explain why net exporting in exchange for gold, would not increase wealth. It would instead only alter prices in the long run.
- These "monetarist" ideas were later developed principally by **Milton Friedman**.

Recap: What is the short run vs. the long run



We are now going to focus on the long run, thus by assumption prices are fully flexible.

Quantity theory of money

Quantity equation (identity):

$$M.V = P.T$$

M = Money supply

V = Velocity of money

- The number of times the average \$ changes hands in a given period of time.

P = Aggregate price level

T = Number of transactions

- The number of times goods and services are exchanged for money.

The nominal value of economic activity is given by how fast can money circulate in the economy and the total quantity of money (currency payments = real transactions)

Quantity theory of money: Assumptions

Assumptions:

1. M is exogenous: Determined outside of the model by the Central bank in the long run.
2. V is constant ($\bar{V}$).
3. T is difficult to measure, therefore replace it with Y (RGDP).
 - $T \uparrow, Y \uparrow$
 - $T \downarrow, Y \downarrow$
4. Y is not a function of M
Depends on fundamentals or the real forces of the economy: preferences and technology.
 $u(C_t), Y = f(K, L)$

Therefore the quantity equation becomes:

$$M \cdot \bar{V} = P \cdot Y$$

How do changes in the money supply impact prices in the long-run?

Quantity equation:

$$M \cdot \bar{V} = P \cdot Y$$

- Since we are interested in thinking about inflation (π), rewrite the quantity equation in terms of growth rates. Taking logs at t and $t - 1$ and using $\ln(1 + g) \approx g$

$$\begin{aligned}\frac{\Delta M}{M} + \frac{\Delta \bar{V}}{\bar{V}} &= \frac{\Delta P}{P} + \frac{\Delta Y}{Y} \\ \Rightarrow \frac{\Delta M}{M} &= \pi + \frac{\Delta Y}{Y} \\ \Rightarrow \pi &= \frac{\Delta M}{M} - \frac{\Delta Y}{Y}\end{aligned}$$

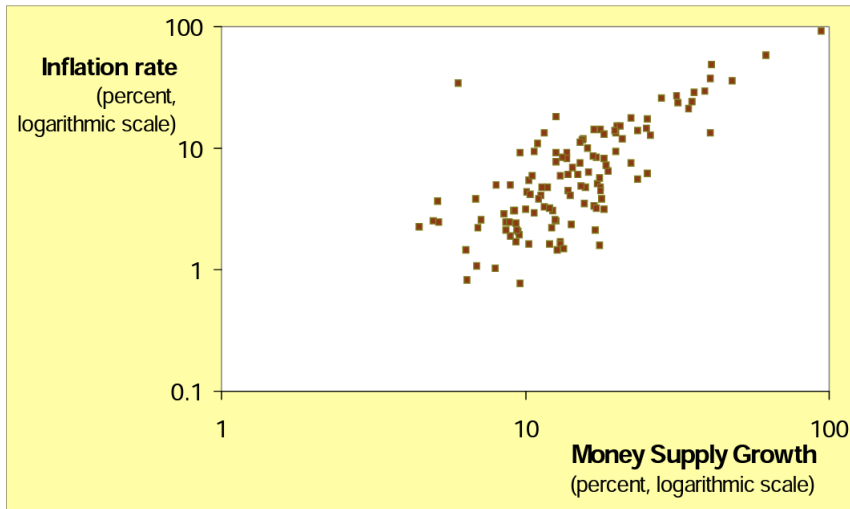
- Since by assumption Y is not a function of M (rules out possible stimulative effect),
 $\Rightarrow$ Any change in M leads to a 1-to-1 change in P (inflation/deflation).

How do changes in the money supply impact prices in the long-run? (2)

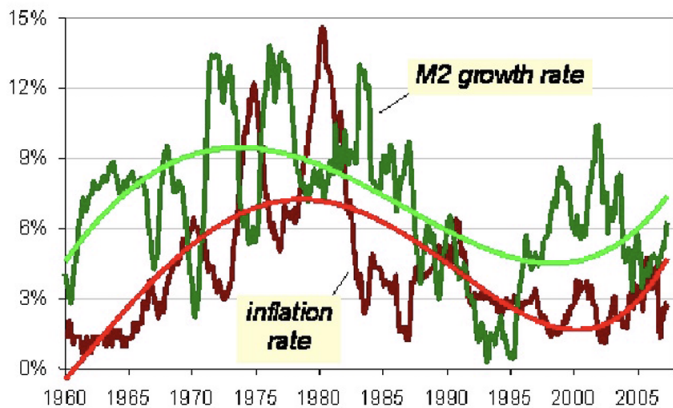
- Monetarist insight:

- *"Inflation is always and everywhere a monetary phenomenon in the sense that it is and can be produced only by a more rapid increase in the quantity of money than in output."* - Milton Friedman.
- **Money is neutral**: Since Y does not depend on M , money supply does not impact real variables (Y), only nominal variables (P).
 - **Classical dichotomy**: in the long run nominal variables (money and prices) are independent of real variables (real GDP, employment, stock of capital...).
- **Money is super-neutral**: Growth rate of Y , does not depend on M .

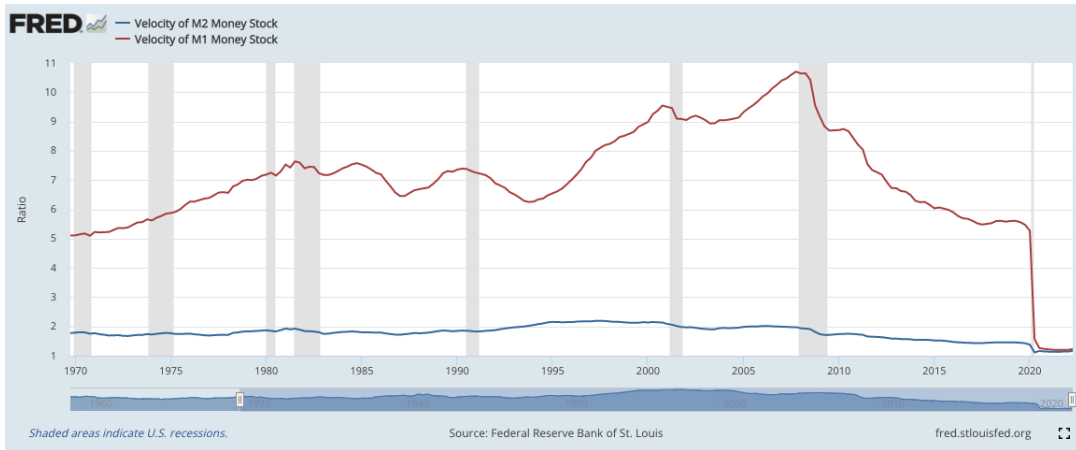
Take it to the data: Long run



Take it to the data: Short run



Is the velocity of money constant?



Interest rates

- Interest rate:
 - **Cost of credit**, i.e. you may borrow to fund an investment.
 - **Opportunity cost of holding money**
 - Think of money as a good that is helpful because it is liquid. (+)
 - But, if you hold cash you are not earning interest. (-)
 - As the interest rate adjusts people's desire to hold cash varies. (interest $\uparrow$, demand for cash $\downarrow$)
- Types of interest rates:
 - **Nominal interest rate (i)**: the rate the bank pays (% of a nominal variable).
 - **Real interest rate (r)**: the rate the bank pays adjusted for inflation, i.e. the change in purchasing power (% of a real variable).

Interest rates and inflation (example)

- Suppose you could buy a one period bond today that has a return of 10% tomorrow. Is it worth it? Is this as good as it seems?
 - Need to account for inflation. What matters is the real rate of return (r).
 - Suppose π^e is 7%, therefore after taking this into account the real rate of return is only 3%.
 - The opportunity is not as good as it seemed, so may no longer be worth it to you.

Fisher equation and Fisher effect

- This example demonstrates:

- Fisher equation:

$$i = r + \pi^e$$

- Fisher Effect : For a given r , a change in inflation expectations (π^e) leads to a 1-to-1 change in i .

⇒ A change in what we expect is going to happen in the future (so nothing has really happened, just our expectations have changed) will change the nominal interest rate today!

Take it to the data

